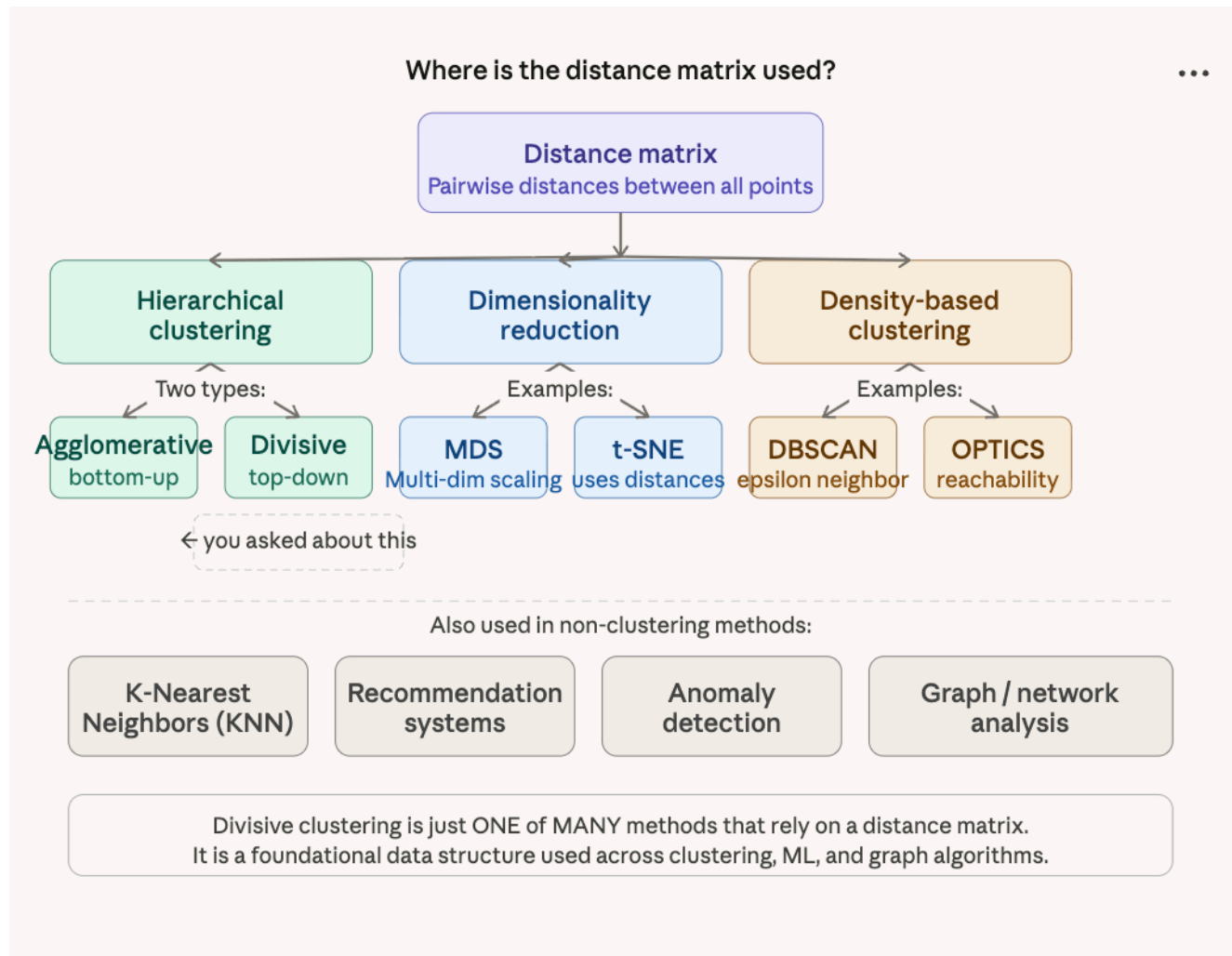


Distance Matrix

Distance Matrix are used in:



While merging two clusters we check the distance between two every pair of clusters and merge the pair with the least distance/most similarity. But the question is how is that distance determined. There are different ways of defining Inter Cluster distance/similarity. Some of them are:

- **Min Distance:** Find the minimum distance between any two points of the cluster.
- **Max Distance:** Find the maximum distance between any two points of the cluster.
- **Group Average:** Find the average distance between every two points of the clusters.
- **Ward's Method:** The similarity of two clusters is based on the increase in squared error when two clusters are merged.



The image compares cluster distance methods:

- Min uses the shortest distance between clusters
- Max uses the longest
- Group Average computes the mean of all pairwise distances
- Ward's method minimizes the increase in within-cluster variance during merging